

Design and Analysis of Algorithm (CS3004-1)

Multiple Choice Questions

Unit-3

- 1` Which of the following is true?
a) **Prim's algorithm initialises with a vertex** b) Prim's algorithm initialises with a edge
c) Prim's algorithm initialises with a vertex which has smallest edge d) Prim's algorithm initialises with a forest
- 2 Prim's algorithm is a _____.
a) Divide and conquer algorithm b) **Greedy algorithm**
c) Dynamic Programming d) Approximation algorithm
- 3 Which of the following is false about Prim's algorithm?
a) It is a greedy algorithm b) **It constructs MST by selecting edges in increasing order of their weights**
c) It never accepts cycles in the MST d) It can be implemented using the Fibonacci heap
- 4 Kruskal's algorithm is used to _____.
a) **find minimum spanning tree** b) find single source shortest path
c) find all pair shortest path algorithm d) traverse the graph
- 5 What is the time complexity of Kruskal's algorithm?
a) $O(\log V)$ b) $O(E^2)$
c) **$O(E \log V)$** d) $O(V \log E)$
- 6 Which of the following is true?
a) Prim's algorithm can also be used for disconnected graphs b) Prim's algorithm is simpler than Kruskal's algorithm
c) In Kruskal's sort edges are added to MST in decreasing order of their weights d) **Kruskal's algorithm can also run on the disconnected graphs**
- 7 Which of the following algorithms is the best approach for solving Huffman codes?
a) exhaustive search b) **greedy algorithm**
c) brute force algorithm d) divide and conquer algorithm
- 8 Dijkstra's Algorithm is used to solve _____ problems.
a) All pair shortest path b) **Single source shortest path**
c) Network flow d) Sorting
- 9 Dijkstra's Algorithm is the prime example for _____.
a) **Greedy algorithm** b) Branch and bound
c) Back tracking d) Dynamic programming
- 10 In Huffman coding, data in a tree always occur?
a) roots b) left sub trees
c) **leaves** d) right sub trees
- 11` What is the primary advantage of the Bellman-Ford algorithm over Dijkstra's algorithm?
a) Works only for positive weights b) **Can handle negative weight edges**

- c) More efficient than Dijkstra's algorithm d) Uses a priority queue
- 12 What is the time complexity of the Bellman-Ford algorithm for a graph with V vertices and E edges?
- a) $O(V)$ b) $O(E)$
c) $O(V^2)$ **d) $O(VE)$**
- 13 If a graph contains a negative-weight cycle, what will Bellman-Ford algorithm do?
- a) Return shortest path normally **b) Terminate and output a warning**
c) Enter an infinite loop d) Ignore the cycle and compute shortest paths
- 14 What is the key principle behind edge relaxation in the Bellman-Ford algorithm?
- a) Updating distances to adjacent vertices if a shorter path is found** b) Assigning random weights to edges
c) Removing unnecessary edges d) Checking for cycles in the graph
- 15 In the state space tree used for backtracking, what do the nodes represent?
- a) The entire solution **b) Partial solutions leading to the final solution**
c) The weight of an edge d) The adjacency matrix
- 16 What is the purpose of the pruning step in backtracking algorithms?
- a) To reduce the number of recursive calls b) To guarantee an optimal solution
c) To avoid checking already visited nodes **d) To eliminate branches that cannot lead to a valid solution**
- 17 In the subset-sum problem, when do we backtrack?
- a) When the sum of the selected elements exceeds the target sum** b) When the sum of the selected elements is equal to the target sum
c) When all elements have been considered d) When a subset has been found
- 18 How does backtracking differ from brute force?
- a) Backtracking does not explore all possibilities **b) Backtracking avoids unnecessary computations by pruning**
c) Backtracking is faster in all cases d) Backtracking guarantees an optimal solution
- 19 In the subset-sum problem using backtracking, what is the worst-case time complexity?
- a) $O(N)$ **b) $O(2^N)$**
c) $O(N \log N)$ d) $O(N^2)$
- 20 What does the Bellman-Ford algorithm detect that Dijkstra's algorithm cannot?
- a) Shortest paths b) Graph cycles
c) Negative-weight cycles d) Directed edges
- 21 Which of the following is the primary purpose of the Branch and Bound method?
- a) To find the exact solution of a problem** **b) To eliminate suboptimal solutions and prune the search space**
c) To provide approximate solutions d) To reduce the problem into smaller sub-

only

problems

- 22 Which of the following is an example of a combinatorial optimization problem solved using Branch and Bound?
- a) Sorting numbers in ascending order
 - b) Multiplying two matrices
 - c) Searching for an element in a list
 - d) **Solving the traveling salesman problem (TSP)**
- 23 In the Job Assignment Problem, what does the objective function typically aim to minimize?
- a) The time to complete all tasks
 - b) The number of tasks assigned
 - c) **The total cost of assigning tasks to workers**
 - d) The total time worked by all workers
- 24 NP-Complete problems are:
- a) Solvable in polynomial time
 - b) Solvable using dynamic programming
 - c) **As hard as the hardest problems in NP**
 - d) Always solvable using Branch and Bound
- 25 In the Knapsack problem, if we are maximizing value and minimizing weight, which method would typically be applied?
- a) **Branch and Bound**
 - b) Greedy method
 - c) Dynamic programming
 - d) Backtracking
- 26 Which of the following statements is true about NP-Hard problems?
- a) They are solvable in polynomial time
 - b) They are more complex than NP-Complete problems
 - c) They cannot be solved in polynomial time, but their solutions can be verified in polynomial time
 - d) **They are at least as hard as NP-Complete problems**
- 27 Which problem is considered NP-Hard?
- a) **Traveling Salesman Problem (TSP)**
 - b) Shortest path problem
 - c) Knapsack problem
 - d) Sorting problem
- 28 Which of the following is NOT a common technique for solving the Job Assignment Problem?
- a) Linear Programming
 - b) **Greedy Algorithm**
 - c) Branch and Bound
 - d) Dynamic Programming
- 29 Which of the following problems is not NP-Complete?
- a) Knapsack Problem
 - b) Subset Sum Problem
 - c) Finding the longest path in a graph
 - d) **Sorting a list of integers**
- 30 Which of the following is the most efficient approach to solve a Knapsack problem with a large number of items?
- a) Brute force search
 - b) Greedy algorithm
 - c) **Dynamic programming or Branch and Bound**
 - d) Backtracking